Shyam Hemamalini



Computational Engineer · Energy & Heat Modelling · Python/Fortran/C++
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Nationality: Dutch | Availability: November 2026 | Project portfolio

About Me

- Computational engineer well-versed in Python, Fortran and C++.
- Graduated with a PhD in Mechanical Engineering from TU Eindhoven, with focus on numerical modelling of turbulent combustion of iron particles.
- Experienced in scientific communication and reporting, with 4 publications and 16 conference presentations.
- Passionate about programming in general, with hobbyist projects in homelabs, web dev and procedural art.
- Looking for a role catering to the energy transition, with focus on building numerical tools and software solutions.

Technical Skills

- **Python Stack** – NumPy, SciPy, pandas, Polars, Matplotlib, h5py (HDF5), Jupyter.
- **Modelling & Analysis** – Heat & mass transfer, time series analysis, scenario & sensitivity analysis, model validation.
- **Scientific Computing** – Numerical methods, HPC, parallelization, data visualization, science communication.
  - **Languages**: Python, HTML/CSS, Fortran, C/C++, MATLAB, CUDA, Bash.
  - **Tools**: Git, Linux, VS Code, Excel, Agentic workflows, OpenFOAM, ParaView.

Education & Experience

PhD in Mechanical Engineering
July 2022 – October 2026
Technische Universiteit Eindhoven — *Eindhoven, Netherlands*

Dissertation Title: Why do iron particles fail to burn? Numerical investigations into the interaction of turbulence and iron powder combustion.

- Developed one-of-the-first numerical frameworks built with Python, Fortran, and C++ to simulate turbulent iron powder combustion.
- Designed and executed simulations at a range of different scales and flow complexities, the results of which are well-cited in the research community.
- Kept the code modular and version-controlled with an Agile-like development process.
- Built a custom, parallel Python post-processing workflow to extract and analyze 500GB+ of output.
- Communicated findings across 16 conferences, and gave 2 invited talks to audiences outside the immediate research group, and won the Best Poster Award at COMBURA 2024.

Research Assistant
Feb. 2022 – Dec. 2022
Wageningen University & Research — *Wageningen, Netherlands*

- Simulated indoor COVID-19 transmission in Ansys Fluent, modelling turbulent airflow and particle transport in multiple configurations.
- Extended Ansys Fluent's solver with a custom, documented C UDF for the flow physics the standard models did not cover.
- Communicated findings on ventilation recommendations to non-technical stakeholders at the university.

MSc Mechanical Engineering – Energy, Flow and Process
Aug. 2019 – Dec. 2021
Technische Universiteit Delft — *Delft, Netherlands*

Master Thesis: A Lagrangian passive scalar solver for mass transport in electrolytes and coupling to the particle-resolved solver Bluebottle.

- Master thesis: Developed a novel C/CUDA/Python quantitative transport model, integrated it into Bluebottle, and validated predictions against experimental data.
- Focus on computational engineering for energy and flow&thermal (F&T) systems, with coursework in thermodynamics, heat and mass transfer, multiphase flow and numerical analysis.
- Teaching Assistant for the edX MOOC *Technology for Intelligent and Integrated Energy Systems* (2021–2023).

Intern – MSc Internship Aug. 2020 – Oct. 2020
TNO - Heat Transfer & Fluid Dynamics group *Delft, Netherlands*

- Modelled multiphase liquid–gas flow and flow-induced vibration in flexible piping.
- Built automated MATLAB pipelines for spectral (PSD) analysis of the time-series data, validated against in-house experiments.

Junior Research Fellow Jul. 2018 – Jun. 2019
PSG College of Technology *Coimbatore, India*

- Developed transient natural-convection models for a heat-sink manifold designed in SolidWorks.
- Optimized the design to 98.5% fin efficiency and validated it experimentally.

Bachelor of Engineering in Mechanical Engineering Aug. 2014 – May 2018
Coimbatore Institute of Technology *Coimbatore, India*

- End-of-degree project: geometrically optimized and fabricated a novel Y-frame folding bicycle, using nonlinear optimization and structural FEA to validate full-vehicle integrity under extreme loads.
- Led CAD/FEA structural validation of an all-terrain vehicle roll cage as Design Engineer, BAJA SAE India.
- Built a line-following robot with a PID-controlled pick-and-place arm for the eYantra Robotics Competition.

Publications

1. *LES of iron-powder combustion in a jet-in-hot-coflow burner – Insights on flame structure and ignition characteristics* 2026
   Proc. Combustion Symposium | Preprint: arXiv:2608.28294
   Shyam Hemamalini, XiaoCheng Mi
2. *Effects of preferential concentration on the combustion of iron particles – A numerical study with homogeneous isotropic turbulence* 2026
   Combustion & Flame | DOI:10.1016/j.combustflame.2026.115119
   Shyam Hemamalini, Benedicte Cuenot, XiaoCheng Mi
3. *A theoretical analysis of timescales in preferential concentration of burning iron particles* 2025
   Fuel - Short Communication (in review) | Preprint: DOI:10.2139/ssrn.5138232
   Shyam Hemamalini, Benedicte Cuenot, XiaoCheng Mi
4. *Numerical study probing the effects of preferential concentration on the combustion of iron particles in a mixing layer* 2024
   Proc. Combustion Symposium | DOI:10.1016/j.proci.2024.105617
   Shyam Hemamalini, Benedicte Cuenot, Jeroen van Oijen, XiaoCheng Mi

Ancillary Activities

Event Photographer 2021 – Present
Part-time professional photographer for various events at TU/e and outside.

Teaching Assistant 2023 – 2025
4RC30 - Introduction to Computational Fluid Dynamics at TU Eindhoven

Teaching Assistant 2021 – 2023
MOOC - "Technology for Intelligent and Integrated Energy Systems" on edX

Volunteer Teacher 2018 – 2019
High-school level physics and chemistry at Abdul Kalam Institute for IIT-JEE, Coimbatore, India

Languages Known

English Fluent
Dutch Intermediate (A2+)
German Intermediate (A2)
French Beginner (A1)
Tamil Native